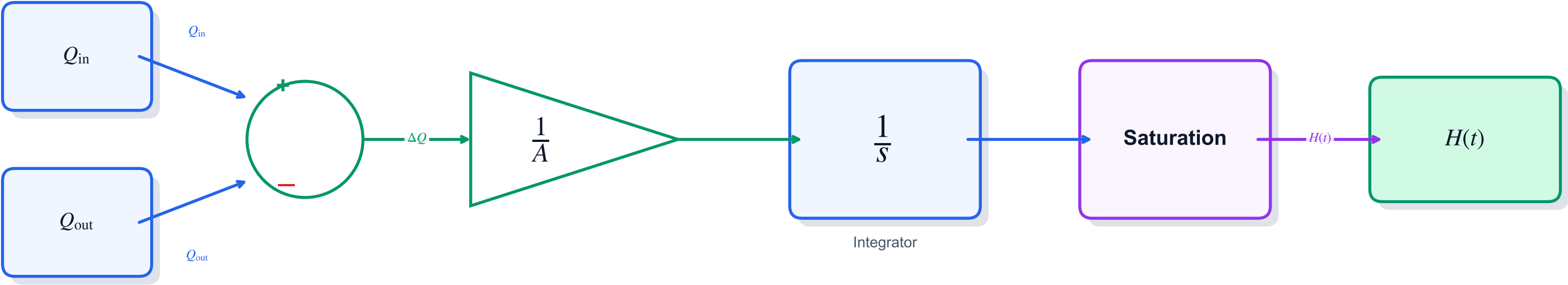


Water Tank Model — Mass Balance Subsystem

$$\frac{dH}{dt} = \frac{Q_{in} - Q_{out}}{A} \quad | \quad H(0) = 4\text{ m} \quad | \quad 0 \leq H \leq H_{max}$$



$$(Q_{in} - Q_{out}) \times \frac{1}{A} \rightarrow \int \rightarrow \text{Sat} \rightarrow H(t)$$

A: Tank Area | IC: 4 m | H_{max} : Max Height